

**Integrated Cross-Species Translation and Biophysical Multi-Scale Modeling Links
Molecular Signatures and Locomotory Phenotypes in Spaceflight-Induced
Sarcopenia**

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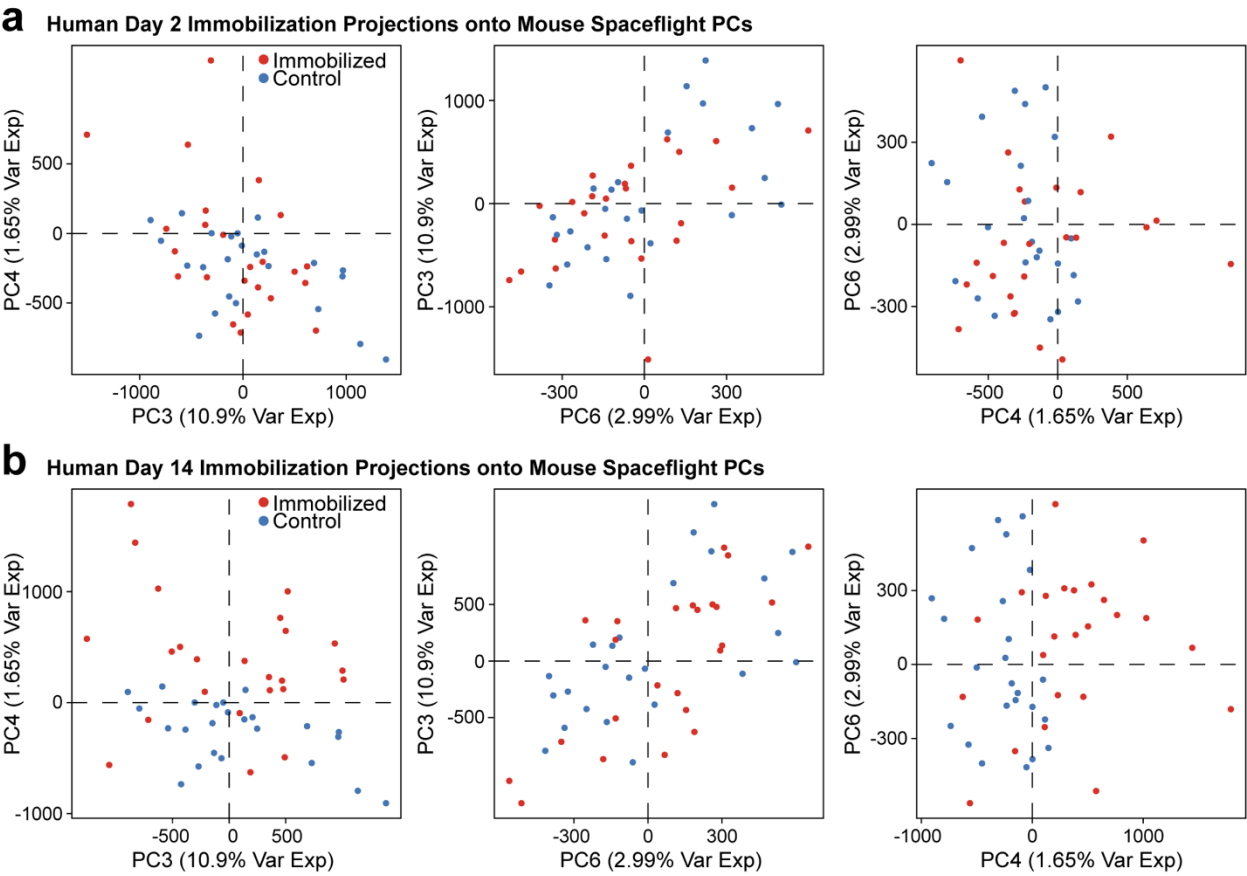
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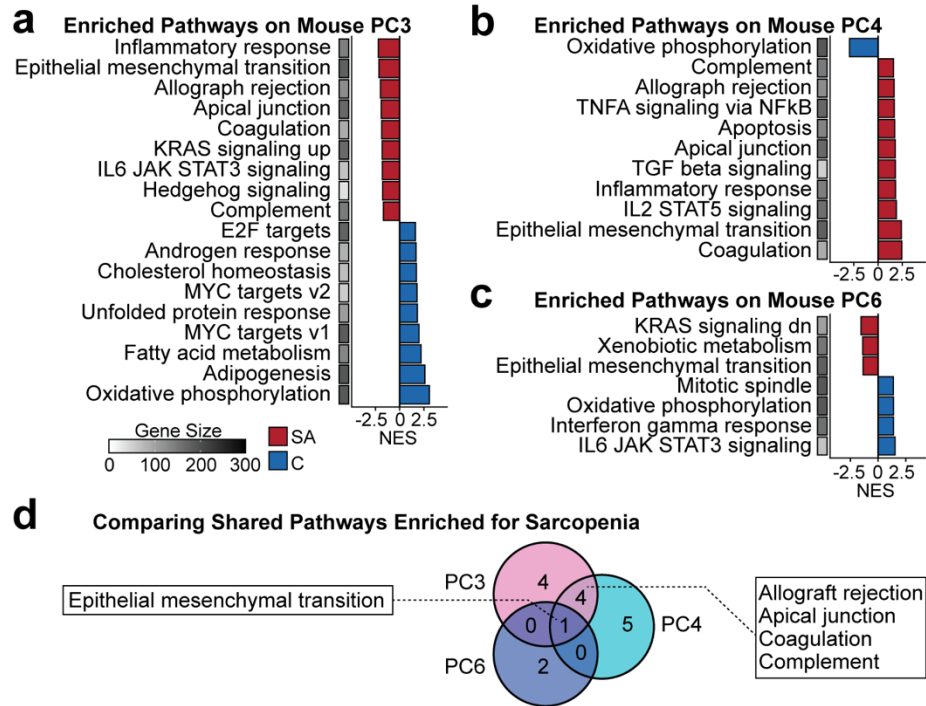
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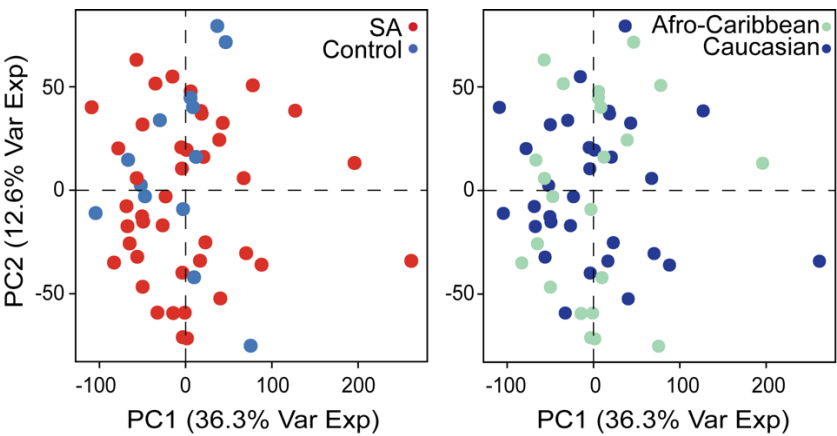
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26 **Supplementary Figure 1.** Plots of human immobilization cohorts projected into the
27 significant mouse spaceflight PCs. **(a)** Human projections comparing day 2 post-
28 immobilization and pre-cast samples. **(b)** Human projections comparing day 14 post-
29 immobilization and pre-cast samples. Sample size (pre-cast = 24 and immobilization =
30 24 per time point. Equal distribution in male and female participants).



Supplementary Figure 2. Hallmark pathway enrichment analysis. (a) Significantly enriched pathways on mouse PC3, (b) PC4, and (c) PC6. (SA=sarcopenia, C=control). (d) Shared sarcopenia-enriched pathways across the PC3, PC4, and PC6. Mouse spaceflight PC3 and PC6 share signatures for sarcopenia, whereas PC4 contain shared signatures for sarcopenia and immobilization-induced atrophy.

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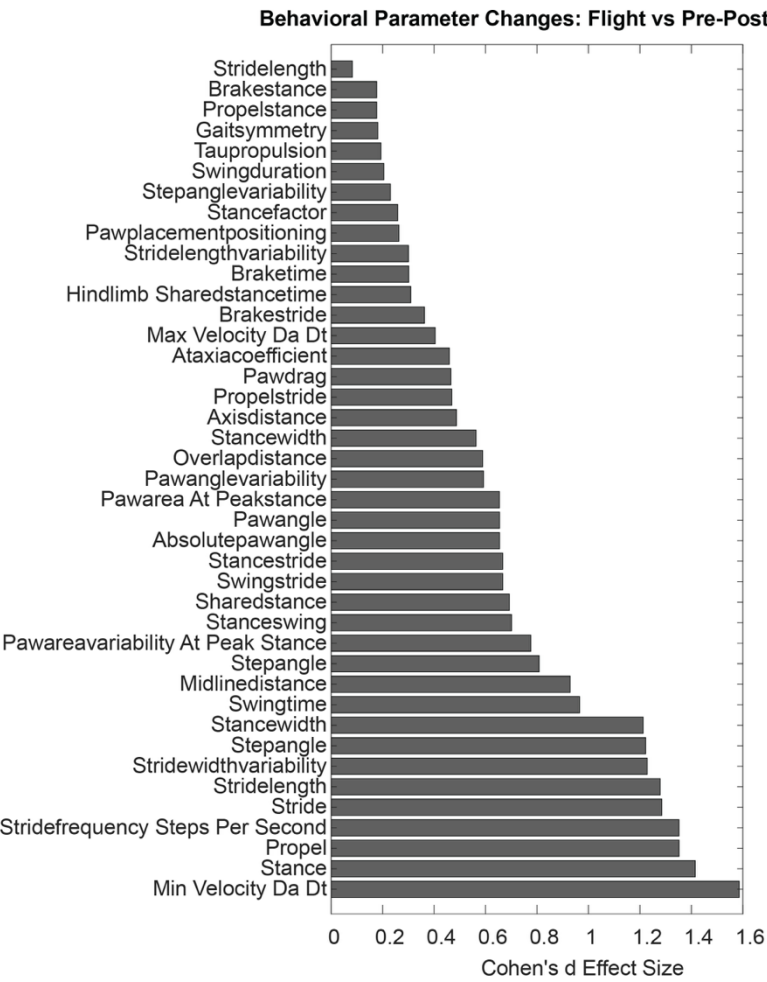
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Supplementary Figure 3. Principal component analysis of the merged Afro-Caribbean and Caucasian dataset. Human patients are labeled by their disease condition and their association with a demographic cohort (SA=sarcopenia).

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Supplementary Figure 4. Effect size of behaviors pre- and post-flight. All gait parameter effect sizes were calculated between pre- and post-flight time points.